

Few-Body Quantum Systems in Harmonic Traps

Summer Research Scholars Report

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August 2026

Abstract

Quantum Monte Carlo (QMC) methods are used in computational nuclear physics to study strongly correlated many-body systems beyond perturbative approximations. We develop a continuous-time quantum Monte Carlo algorithm for a few fermions in three dimensions confined by a harmonic oscillator potential. Fermions are sampled through hard-core bosonic worldline configurations, with antisymmetry recovered by measuring the permutation sign. The propagator $e^{-\beta H}$ is expanded in hopping events, and each sweep proposes local insertions, removals, and shifts of these events. The model provides a controlled testbed for importance sampling, parallel Markov-chain averaging, finite-temperature convergence, density reconstruction, and the fermion sign problem. Simulations use fixed-particle-number sectors, open or periodic boundaries, and independent chains. Observables include total energy, site occupation, acceptance rates, and the average sign. Three-dimensional occupation maps and radial density profiles show how harmonic confinement redistributes particles between a dense center and lower-density shells. The algorithm is benchmarked against single-particle solutions and exact diagonalizations on small lattices.

1 Research Questions and Project Goals

QMC methods are the most practically advantageous for studying strongly correlated quantum systems. Thus we wanted to see how can a Monte Carlo algorithm be constructed that samples the imaginary-time evolution of a small quantum system while remaining flexible enough to study bosons, fermions, external trapping potentials, and multiple spatial dimensions. The harmonic oscillator was chosen as a particularly useful test problem because its single-particle physics is well understood, while the few-body lattice problem still exposes the numerical issues that appear in more complicated many-body calculations.

The work was organized around five related research questions:

1. **Algorithmic validation.** Can a continuous-time/operator-expansion QMC algorithm reproduce exact finite-lattice energies for small fixed- N systems?
2. **Fermionic antisymmetry.** Can fermions be sampled through positive hard-core-boson worldline weights while recovering the correct fermionic observables through permutation-sign reweighting?
3. **Convergence and computational scaling.** How do inverse temperature β , operator-string cutoff M , particle number N , boundary conditions, the number of Monte Carlo sweeps, and independent-chain averaging affect the accuracy and stability of the calculation?

4. **Harmonic confinement.** How does an external potential

$$V_i = \frac{1}{2}m\omega^2|\mathbf{r}_i - \mathbf{r}_c|^2 \quad (1)$$

redistribute density on a finite lattice, particularly in three dimensions?

5. **Toward many-body structure.** As particles are added to a trap, can site occupations and radial profiles distinguish a dense central region from a partially occupied outer shell, and what additional observables would be required to determine whether that shell is coherent?

A secondary goal was to build a reusable computational framework rather than a one-off calculation. This led to substantial work on modular code, parallel execution, diagnostic output, exact checks, data export, and post-processing.

2 Theoretical Framework

2.1 Lattice Hamiltonian and fixed-particle-number states

The physical model consists of particles hopping between nearest-neighbor sites with an optional onsite trapping potential. In its simplest form,

$$H = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i) + \sum_i V_i n_i, \quad n_i = c_i^\dagger c_i. \quad (2)$$

For the hard-core problem, $n_i \in \{0, 1\}$, and the simulations remain in a fixed sector

$$\sum_i n_i = N. \quad (3)$$

The same occupation basis can therefore represent hard-core bosons or spinless fermions. Their difference appears in the sign acquired when identical fermions are permuted.

The harmonic trap is centered at $\mathbf{r}_c$, normally the geometric center of the finite lattice. For a cubic $L \times L \times L$ lattice with spacing a ,

$$\mathbf{r}_i - \mathbf{r}_c = a(x_i - x_c, y_i - y_c, z_i - z_c), \quad (4)$$

so the trap strength can be varied through m , ω , and a . Open boundaries are physically natural for a trap, while periodic boundaries remain useful for benchmark calculations because the free lattice spectrum is known analytically.

2.2 Continuous-time/operator-expansion Monte Carlo

The starting point is the imaginary-time propagator

$$e^{-\beta H}. \quad (5)$$

The theoretical derivation begins by dividing β into short intervals and expanding the propagator in local bond operators. In the continuous-time limit, a configuration C can be viewed as an ordered set of local operators placed along the imaginary-time direction,

$$0 < \tau_1 < \tau_2 < \cdots < \tau_{N_{\text{op}}} < \beta. \quad (6)$$

The configuration weight is the product of the corresponding local matrix elements. Detailed-balance ratios for inserting or removing a diagonal operator can be written so that the auxiliary time step cancels. This is the practical reason the method can be interpreted in the continuous-time limit rather than as a calculation with a physical Trotter time-step error.

The implementation stores the expansion in an operator string with maximum length M . In this context, M is a storage and truncation parameter, not the number of physical imaginary-time slices. A reliable run requires the sampled expansion order to remain comfortably below M ; increasing β , the trap strength, the lattice size, or the positive operator shifts can all increase the typical expansion order and therefore require a larger M .

The principal updates explored during the project were:

- **Diagonal insertion/removal:** add or remove a diagonal bond or onsite-potential operator using Metropolis probabilities derived from detailed balance.
- **Worm/vertex conversion:** convert compatible diagonal vertices into off-diagonal hopping vertices and propagate changes through the operator string while respecting worldline continuity.
- **Pair-hop and state updates:** propose additional changes that move occupation while preserving the fixed total particle number.

These moves are complementary. Diagonal updates change the expansion order, while worm-like updates are necessary to change worldline topology and sample hopping configurations efficiently.

2.3 Energy estimator and positive shifts

For an operator expansion whose configuration weight contains a factor $\beta^{N_{\text{op}}}$, differentiating the partition function with respect to β gives the familiar expansion-order estimator

$$E_{\text{exp}} = -\frac{\langle N_{\text{op}} \rangle}{\beta}. \quad (7)$$

The implementation adds diagonal constants to local operators so that all sampled diagonal matrix elements are nonnegative. These shifts do not change eigenstates, but their known contribution must be added back to obtain the physical energy.

The harmonic potential required special care because directly sampling $-V_i n_i$ in the expansion can produce negative diagonal weights. The code instead chooses a positive constant $C_i > V_i$ and rewrites

$$H_V = \sum_i C_i - \sum_i (C_i - V_i n_i). \quad (8)$$

The sampled onsite matrix element $C_i - V_i n_i$ is then positive, while $\sum_i C_i$ is restored in the final energy estimator. The same bookkeeping principle is used for the constant shifts in the kinetic bond operators.

2.4 Fermion sign reweighting

For hard-core bosons, configuration weights can be sampled as ordinary nonnegative probabilities. For fermions, a worldline configuration can acquire a minus sign whenever particle trajectories produce an odd permutation. If $P(C)$ is the permutation parity,

$$s(C) = (-1)^{P(C)}. \quad (9)$$

The code therefore samples the absolute weight $|W(C)|$ and measures the sign as part of the observable. A fermionic expectation value is obtained from

$$\langle O \rangle_F = \frac{\langle sO \rangle_{|W|}}{\langle s \rangle_{|W|}}. \quad (10)$$

For the expansion-order contribution to the energy,

$$E_F = -\frac{1}{\beta} \frac{\langle sN_{\text{op}} \rangle_{|W|}}{\langle s \rangle_{|W|}} + \text{known shift corrections}. \quad (11)$$

This approach is exact in principle, but it exposes the fermion sign problem directly: when $\langle s \rangle$ becomes very small, the denominator in Eq. (10) is determined by a near-cancellation of positive and negative samples, and statistical errors grow rapidly.

3 Computational Methodology and Code Development

3.1 Prototype and exact checks

The first implementation was a Python prototype for a small periodic lattice. It explicitly generated all hard-core basis states with fixed N , represented each operator as either an identity, diagonal bond, or hopping bond, propagated a state through the operator string, and rejected strings that failed to close in imaginary time. The prototype separated thermalization sweeps from measurement sweeps and recorded acceptance rates for diagonal, toggle, pair-hop, and initial-state updates.

A key feature of the prototype was an exact finite-temperature comparison. The kinetic Hamiltonian was built explicitly in the fixed- N basis, diagonalized with standard linear algebra, and used to calculate

$$E(\beta) = \frac{\sum_n E_n e^{-\beta E_n}}{\sum_n e^{-\beta E_n}}. \quad (12)$$

This made small systems an immediate unit test for the Monte Carlo sampler and helped distinguish sampling errors from physics errors.

3.2 C++ production implementation

The production code was then moved to C++ for performance. The implementation supports:

- hard-core bosons and spinless fermions;
- fixed particle number N ;
- open or periodic boundaries;
- tunable L , β , t , M , thermalization sweeps, measurement sweeps, and measurement interval;
- harmonic confinement with configurable m , ω , a , and trap center;
- energy, expansion-order, hopping/diagonal operator counts, potential energy, average sign, sign error, and update acceptance rates;
- site-resolved occupations $\langle n_i \rangle$;

- CSV export and interactive Plotly occupation maps.

The code was first used on square lattices and then generalized to a true cubic lattice with L^3 sites. For example, $L = 4$ corresponds to 64 sites in the three-dimensional version. The three-dimensional visualization records the site coordinates, physical coordinates relative to the trap center, onsite potential, occupation, normalized one-particle probability, chain count, and number of pooled measurements.

3.3 Parallel Markov chains

Long QMC runs are naturally parallelized by running statistically independent chains. The production workflow uses OpenMP to execute up to 12 chains simultaneously, matching the 12-core computing environment used during the project. Each chain is given a distinct seed, thermalizes independently, and returns raw measurement accumulators.

A particularly important implementation detail was how chain results were combined. For bosons, averaging occupations across all measurements is straightforward. For fermions, it is incorrect in general to average already-divided per-chain quantities when the chains have different sign sums. Instead, the pooled estimator is formed from raw signed sums,

$$\langle n_i \rangle_F = \frac{\sum_c \sum_m s_{cm} n_{i,cm}}{\sum_c \sum_m s_{cm}}. \quad (13)$$

This post-aggregation design also prevents each chain from writing a separate map file. The program writes one CSV and one interactive map only after all chains have been combined.

The parallel version was run both locally and on a Jetstream2 virtual machine. Typical high-statistics parameter sets used 10^5 thermalization sweeps and up to 10^6 measurement sweeps per chain. Very long runs demonstrated that raw sweep count alone is not an efficient solution to every convergence problem, particularly once the fermion sign becomes small.

3.4 Uncertainty estimation and diagnostics

Because consecutive configurations in a Markov chain are correlated, a naive standard error from individual measurements is too optimistic. The project therefore used blockwise accumulation and examined reblocking/bootstrap logic as a way to estimate uncertainty after correlations have been reduced. In addition to the final energy error, the code reports quantities that are useful for diagnosing a run before trusting the physics:

- update acceptance rates;
- average expansion order and operator-type counts;
- fraction of samples that approach the cutoff M ;
- average sign and sign uncertainty;
- occupation normalization $\sum_i \langle n_i \rangle = N$;
- agreement among independent chains.

4 Results

4.1 Hard-core-boson benchmark against exact results

The cleanest validation was the periodic 4×4 hard-core-boson problem at $\beta = 1$ and $t = 1$. The QMC values closely reproduce the exact lattice results across multiple particle-number sectors. Table 1 reproduces the benchmark set from the project notes. The parenthetical notation in the original calculation has been written here as an explicit one-standard-error uncertainty.

Table 1: Periodic 4×4 hard-core-boson benchmark at $\beta = 1$ and $t = 1$. Energies use the shifted lattice-Hamiltonian convention employed by the QMC code.

N	QMC energy	Exact energy
2	-63.573 ± 0.004	$-63.569\,14$
3	-62.663 ± 0.005	$-62.663\,98$
4	-61.255 ± 0.005	$-61.250\,74$
5	-59.294 ± 0.004	$-59.299\,42$
6	-56.787 ± 0.004	$-56.787\,04$
7	-53.693 ± 0.004	$-53.692\,63$
8	-49.999 ± 0.004	$-49.998\,06$

This agreement establishes several things at once: the local operator rules are consistent, the update sequence reaches the correct equilibrium distribution, the expansion-order energy estimator is normalized correctly, and the fixed- N bookkeeping is functioning over a range of fillings.

4.2 Fermionic benchmark and the sign problem

The same framework can be used for fermions by sign reweighting, but the statistical difficulty changes dramatically. Table 2 shows the $\beta = 1$ benchmark. At $N = 2$, the average sign is still roughly 0.13 and the QMC result agrees well with the exact value. By $N = 4$ and $N = 5$, the average sign is of order 10^{-3} to 10^{-2} and the uncertainty grows sharply. At $N = 8$, the quoted uncertainty is several energy units even though the underlying sampler is still running.

Table 2: Spinless-fermion benchmark on the periodic 4×4 lattice at $\beta = 1$. The deterioration of precision follows the collapse of the average sign.

N	$\langle s \rangle$	QMC energy	Exact energy
2	0.128 39	-61.869 ± 0.007	-61.864
3	0.037 51	-59.857 ± 0.027	-59.861
4	0.005 34	-57.302 ± 0.214	-57.107
5	0.002 46	-55.137 ± 0.458	-55.593
8	0.002 46	-44.841 ± 2.964	-43.488

A higher-temperature test, $\beta = 0.2$, produced much healthier signs. For example, $\langle s \rangle = 0.59523$ for $N = 2$ and remained around 0.1 for many of the larger sectors. The resulting energies tracked exact values closely. Figure 1 summarizes the contrast. This was one of the most important physical

and computational lessons of the project: increasing β is desirable for ground-state projection, but it simultaneously makes the fermionic estimator exponentially more expensive because positive and negative configurations cancel more strongly.

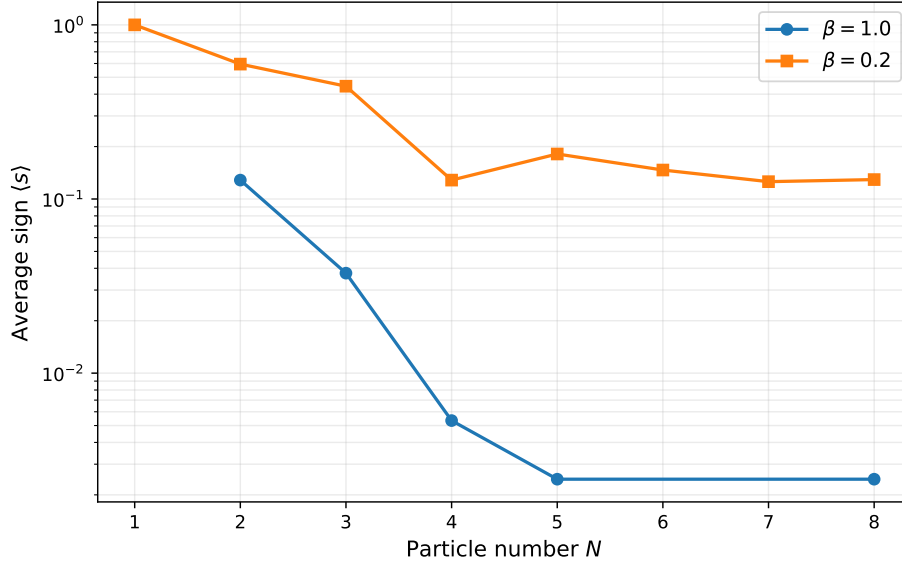


Figure 1: Average fermion sign versus particle number for two inverse temperatures from the benchmark calculations. The lower-temperature calculation ($\beta = 1$) shows much more severe cancellation.

Additional sign-structured tests, including a π -flux variant of the hopping problem, were used as stress tests for the fermion estimator. They reinforced the same trend: agreement with exact diagonalization is straightforward when the sign remains appreciable and progressively more expensive as the sign collapses.

4.3 Harmonic confinement and density reconstruction

The harmonic-trap extension was first checked on small systems where exact diagonalization was still feasible. For a periodic 4×4 hard-core-boson system with $N = 2$, $m = 1$, $\omega = 0.5$, $a = 1$, and $t = 1$, exact diagonalization of the same lattice Hamiltonian gave a ground-state energy

$$E_0 = -62.9672420519. \quad (14)$$

More importantly for validating the density observable, the occupations sum to exactly 2 and become largest at the four central sites. The central-site occupation is approximately 0.148518, while the corner occupation is approximately 0.103250. Without the trap, the same periodic $N = 2$ system has the uniform occupation $2/16 = 0.125$ on every site. Figure 2 shows the resulting redistribution.

The large negative numerical energies in Tables 1–2 and in the trapped benchmark should not be compared directly with the familiar continuum oscillator spectrum $(n + 1/2)\hbar\omega$. They include the explicit constant shifts used in the lattice-QMC operator decomposition. The meaningful validation is QMC versus exact diagonalization under the same Hamiltonian convention, together with the correctly reconstructed physical density.

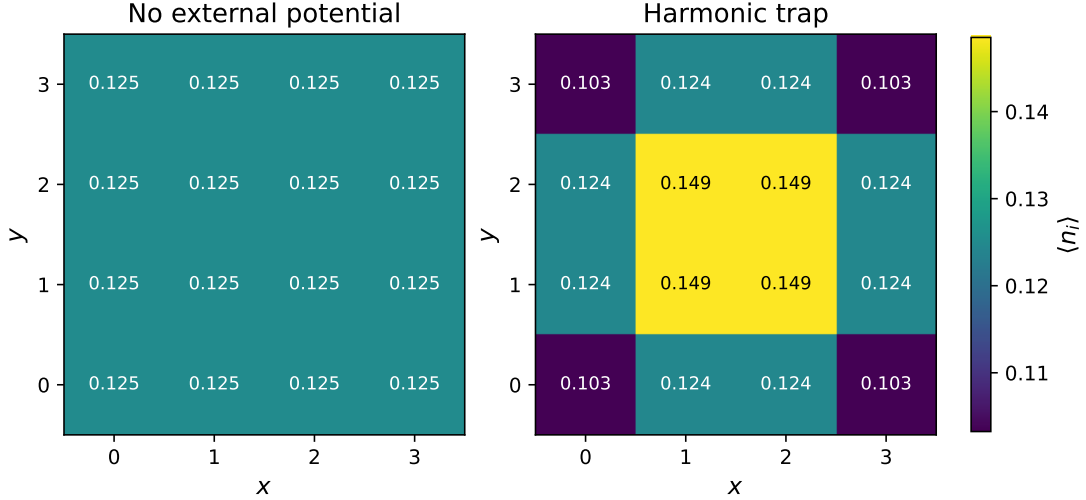


Figure 2: Exact small-lattice density benchmark for $N = 2$. The untrapped periodic system is uniform, while the harmonic potential transfers occupation toward the center. These exact data were used as a reference for the QMC density estimator.

4.4 Three-dimensional occupation maps and shell analysis

The next stage generalized the calculation to a cubic L^3 lattice and produced site-resolved three-dimensional occupation maps. A short 12-chain validation run confirmed the aggregation logic: 12 chains with 20 measurements each yielded one combined map with 240 total measurements and an occupation sum equal to the fixed particle number $N = 2$. Production configurations were then designed for substantially longer runs, commonly using 10^5 thermalization sweeps and 10^6 measurement sweeps per chain.

To make the three-dimensional density physically interpretable, the post-processing pipeline groups sites into radial shells according to equal values of

$$r_i^2 = x_i^2 + y_i^2 + z_i^2 \quad (15)$$

relative to the trap center. For a shell containing N_r lattice sites, the radial density is

$$n(r) = \frac{1}{N_r} \sum_{i \in r} \langle n_i \rangle. \quad (16)$$

The code also compares consecutive particle-number sectors through

$$\Delta_N n_i = \langle n_i \rangle_{N+1} - \langle n_i \rangle_N, \quad (17)$$

with an analogous radially averaged $\Delta_N n(r)$. These quantities answer a more specific question than a single density map: once central sites approach unit occupation, where does the *next* particle enter the trap? A shift of $\Delta_N n(r)$ toward larger r would be the expected signature of a developing outer shell.

The shell-study code was also modified to avoid explicit enumeration of every fixed- N basis state. The basis size grows combinatorially, and that representation becomes a memory limitation well before a local worldline sampler itself becomes impossible. The basis-free version instead uses symmetric fixed- N proposals and local particle-hole moves. This removes the explicit basis-memory bottleneck, although acceptance and equilibration still need to be checked carefully as N grows.

At the end of the summer, the occupation and radial-density machinery was in place, but the shell analysis remained a convergence-sensitive next step rather than a claim of a fully characterized phase. In particular, the current radial-profile plotting script uses the spread among symmetry-equivalent sites as a visual error bar; statistically rigorous Monte Carlo uncertainty on $n(r)$ will require storing blockwise site-occupation estimators in the C++ output.

4.5 Computational lessons

Several practical lessons emerged from implementation and testing:

- **Independent seeds matter.** Parallel chains must not reuse identical random-number streams. Distinct chain seeds were added before pooling results.
- **M must scale with the problem.** Increasing β , lattice size, trap curvature, mass, or lattice spacing increases the expected expansion order. A cutoff that is safe for a free $\beta = 1$ run may not be safe for a stronger or colder trapped calculation.
- **More sweeps cannot cure a vanishing sign efficiently.** Runs with tens of millions of sweeps become computationally expensive, while the variance of a sign-reweighted estimator can still dominate.
- **Occupation normalization is a powerful diagnostic.** For a fixed- N calculation, $\sum_i \langle n_i \rangle = N$ provides an immediate check on estimator and aggregation logic.
- **Exact diagonalization remains essential.** Although it scales poorly, exact calculations for $N = 1$ or small L, N are invaluable for validating energy, density, boundary conditions, and trap conventions before larger QMC runs are trusted.

5 Planned and Potential Next Steps

The project produced a working framework, but several natural extensions would turn the present calculations into a more complete study of trapped few-body physics.

5.1 Systematic convergence study

The first priority is to complete controlled one-factor-at-a-time scans in M , L , β , and the number of measurement sweeps. The shell-study driver was set up with representative scans such as

$$M \in \{800, 1200, 1600\}, \quad \beta \in \{1, 2, 4\}, \quad N_{\text{sweeps}} \in \{2.5 \times 10^4, 10^5, 4 \times 10^5\}.$$

The goal is to separate four effects that can otherwise be confused: operator-cutoff error, finite-temperature error, finite-volume/lattice-spacing error, and ordinary Monte Carlo uncertainty.

A particularly important continuum-oriented test is to increase L while reducing a , holding the physical trap scale fixed. This would clarify which density structures are genuine physical features and which are artifacts of a coarse lattice.

5.2 Consecutive- N shell formation

The radial and $\Delta_N n_i$ analysis should be run over a sequence of fixed-particle-number sectors. A useful target is to start with $N = 1$ through 8 on a 4^3 lattice and then increase N gradually only after checking cutoff frequency, acceptance rates, occupation normalization, and chain-to-chain agreement. The resulting $n(r)$ and $\Delta_N n(r)$ curves would determine whether added particles transition from filling the center to predominantly occupying an outer shell.

5.3 Blockwise density uncertainties

The energy and sign already have blockwise uncertainty machinery, but site occupations should be stored block-by-block as well. That would allow bootstrap or jackknife estimates for $\langle n_i \rangle$, $n(r)$, and $\Delta_N n(r)$ and would distinguish Monte Carlo noise from the spread among symmetry-related lattice sites.

5.4 One-body density matrix and momentum distribution

Density alone cannot determine whether a partially occupied shell is coherent. The most important physics extension is therefore to measure the one-body density matrix

$$\rho_{ij} = \langle b_i^\dagger b_j \rangle. \quad (18)$$

A worm estimator is well suited to this observable because the open ends of the worm naturally sample off-diagonal propagation. Once ρ_{ij} is available, the momentum distribution can be reconstructed as

$$n(\mathbf{k}) = \sum_{ij} e^{i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \rho_{ij}. \quad (19)$$

This would allow the project to distinguish a dense but incoherent shell from one with substantial off-diagonal coherence.

5.5 Interactions and two-component fermions

A longer-term extension is to introduce two fermion species, $\sigma = \uparrow, \downarrow$, and an onsite interaction,

$$H_{\text{int}} = U \sum_i n_{\uparrow, i} n_{\downarrow, i}, \quad (20)$$

with the corresponding diagonal interaction vertices connecting the two worldline layers. This would move the calculation closer to the interacting few-fermion systems of interest in cold atoms and nuclear lattice calculations. Before that step, however, the noninteracting trapped problem should be fully converged and the sign scaling quantitatively characterized.

6 Conclusion

The summer project developed a functioning QMC framework from first principles and used it to explore both the promise and the limitations of worldline Monte Carlo for few-body quantum systems. The hard-core-boson calculation reproduces exact small-lattice energies to within Monte Carlo uncertainty. The

fermion implementation recovers antisymmetry through permutation-sign reweighting and reproduces exact results while the average sign remains sufficiently large, but it also demonstrates directly how statistical precision collapses as the sign approaches zero. The harmonic-trap extension adds a physically motivated inhomogeneous potential while maintaining positive sampling weights through diagonal shifts, and small-lattice exact calculations confirm the expected transfer of density toward the trap center.

The project also produced a practical three-dimensional research code: it supports open and periodic boundaries, fixed- N sectors, 12-chain parallel execution, sign-aware aggregation, occupation maps, exact checks, radial density analysis, and consecutive-sector comparisons. The most promising next step is to use this framework to characterize the formation of a dense core and outer shell in a three-dimensional trap, then extend the worm estimator to the one-body density matrix so that density and coherence can be studied separately. In that sense, the project progressed from validating an algorithm to establishing the computational infrastructure for a broader study of trapped few-body quantum matter.

References

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